

Ha Do

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EDUCATION

The University of Chicago

September 2022 — June 2026

B.S. in Astrophysics & B.A. in Race, Diaspora, and Indigeneity

Cumulative GPA: 3.88/4.00

Relevant Coursework

General Physics, Mathematical Methods for the Physical Sciences; Python Programming with Applications to Astrophysics; Computational Techniques in Astrophysics, Observational Techniques in Astrophysics, Undergraduate Research Seminar, Quantum Mechanics, Astrophysics Field Course, Radiation Processes, Statistical & Thermal Physics, Physics of Stars, Physics of Galaxies (to be completed)

SCHOLARSHIPS, AWARDS, & HONORS

Odyssey Scholar | Davis UWC Scholar These awards provide full funding for undergraduate degree

Department of Race, Diaspora, and Indigeneity Writing Prize Spring 2025

Quad Research Scholar 2024-2026 | Straetz International Research Scholar 2025 (declined)

Dean's list 2022-2023, 2023-2024 Top 20% GPA in the College

Robert Maynard Hutchins Scholar 2023-2024 Top 10% GPA in graduating class

RESEARCH EXPERIENCE

University of Chicago, Exploring the chemical evolution of Palomar 5

Undergraduate Researcher, September 2025 – Present

Advisors: Alexander Ji

Analyse spectra for chemical abundances of stars in the Palomar 5 stellar stream to reconstruct its star-formation history, production of neutron-capture elements, and place it within the broader framework of hierarchical galaxy assembly.

University of Chicago, Tracing the origins of r-process nucleosynthesis in Reticulum II

Undergraduate Researcher, September 2025 – Present

Advisors: Alexander Ji

Honours thesis project: Analyse the chemical composition of a particularly metal-poor star in the dwarf galaxy Reticulum II, especially for detections of Zn as signals of r-process nucleosynthesis from core-collapse supernovae.

UC Berkeley, Increasing Efficiency of Target Selection for SETI

Undergraduate Researcher, June 2025 – August 2025

Advisors: Daniel Czech, Chenoa Tremblay, Brian Lacki

Developed target selector and prioritisation code for SETI observations using commensal observing techniques on the MeerKAT and VLA arrays.

University of Chicago, Analysis of the Most Pristine Star

Undergraduate Researcher, January 2025 – June 2025

Advisors: Alexander Ji

Collaboratively identified, observed, and analysed a uniquely metal-poor, chemically pristine star using Magellan/MIKE spectroscopy, with group findings under review in *Nature Astronomy*.

University of Chicago, Quasar Analysis over a Century

Undergraduate Researcher, January 2025 – Present

Advisors: Richard Kron

Digitized historic glass plates from the Yerkes Observatory to recover photometric data of quasars down to nearly the 19th magnitude in order to analytically compare to modern SDSS photometry.

University of Chicago, Investigating the Jet Stream Using the DECam MAGIC Survey

Undergraduate Researcher, December 2023 – Present

Advisors: Anirudh Chiti, Alexander Ji

Led the analysis of the Jet Stream's spatial, motion, and metallicity distributions, competitively obtaining NSF NOIRLab telescope time with CTIO/DECam as co-PI to map its full extent. First-author manuscript in preparation for submission to *The Astrophysical Journal*.

University of Chicago, Analysis of Spectrograms from ϵ Aurigae's 1928-1930 Eclipse

Undergraduate Researcher, October 2022 – January 2025

Advisors: Richard Kron

Developed digitization methods to analyse historic spectrograms of the 1928-1930 eclipse of ϵ Aurigae. Analyzed scientific data from digitisations, including spectra calibration, photometric calibration, and astrometric calibration. Third author of paper in *Publications of the Astronomical Society of the Pacific*.

PUBLICATIONS, PRESENTATIONS, & CONFERENCES

Ha Do, Anirudh Chiti, Peter S. Ferguson, Alexander P. Ji, Guilherme Limberg, T. S. Li, Alex Drlica-Wagner, Andrew B. Pace, David Sand, Kaia Atzberger, Vinicius Placco, Jeffrey Carlin, Noelia E. D. Noel, *The DECam MAGIC Survey: Investigating the Morphology of the Jet Stream*. Under collaboration-wide review for submission to **ApJ**.

Alexander P. Ji, Vedant Chandra, Selenna Mejias-Torres, Zhongyuan Zhang, Philipp Eitner, Kevin C. Schlaufman, Hillary D. Andales, **Ha Do**, Natalie M. Orrantia, Rithika Tudmilla, Pierre N. Thibodeaux, Keivan G. Stassun, Madeline Howell, Jamie Tayar, Maria Bergemann, Andrew R. Casey, Jennifer A. Johnson, William Cerny, Keith Hawkins, Guilherme Limberg, Guy S. Stringfellow, Riley Thai, *A Nearly Pristine Star from the Large Magellanic Cloud*. 2026, *Nature Astronomy*, accepted. [Preprint](#)

Escapa I., Babnigg D., **Do H.**, Kovach-Fuentes R., Kron R., Millan Ortiz N., *Digitization and Evaluation of Spectrograms from ϵ Aurigae's 1928-1930 Eclipse*. 2025, **PASP**, 137, 014202. [Publication](#)

University of Chicago Undergraduate Research Symposium, Chicago, IL, April 2025. **Ha Do**, Anirudh Chiti, Alexander Ji. *Investigating the Jet Stream using the DECam MAGIC Survey*. (Poster, Presenter)

Dwarf Galaxies, Star Clusters, and Streams in the LSST Era Workshop, Kavli Institute of Cosmological Physics, Chicago, IL, June 2024 (Attendee).

244th Meeting of the American Astronomical Society, Madison, WI, June 2024. Daniel Babnigg, **Ha Do**, Nicole Millan Ortiz, Sandra Schloen, Richard Kron. *Edwin Hubble's Doctoral Thesis: Using OCHRE to Associate Interdisciplinary Materials* (iPoster, Presenter).

243rd Meeting of the American Astronomical Society, New Orleans, LA, January 2024. Isaiah Escapa, Daniel Babnigg, **Ha Do**, Rachel Kovach-Fuentes, Harper Learmonth, Nicole Millan Ortiz, Ava Trent, Richard Kron. "Digitization and Analysis of Spectrograms of Epsilon Aurigae from the 1929 Eclipse" (iPoster).

Biennial History of Astronomy Workshops, The University of Notre Dame, Notre Dame, IN, June 2023. Babnigg, Daniel; **Do, Ha**; Kovach-Fuentes, Rachel. "Digitization and Analysis of a Yerkes Observatory Spectroscopic Glass Plate" (Paper Session, Presenter).

OBSERVING EXPERIENCE & ALLOCATIONS

Observer training, Allen Telescope Array — Radio interferometry; on-site

Co-PI, 2.5 nights on CTIO/DECam (2025A) — Imaging; remote & on-site; **competitively obtained**

Observer, 3 nights on Magellan/MIKE — Single-slit spectroscopy; remote & on-site

Observer training, 0.5 nights on Magellan/MagE — Single-slit spectroscopy; remote

Observer, 4 nights on Stone Edge Observatory — Imaging; remote

TEACHING EXPERIENCE

The University of Chicago, ASTR 13300, Introduction to Astrophysics | College level

Teaching Assistant, Spring 2025

Provided guidance for students in a class of 46 during office hours, covering introductory astrophysics topics. Graded and corrected coursework.

The University of Chicago, ASTR 11901 Physics of Stars: An Introduction x 2 | High school level

Teaching Assistant, July 2023 & July 2024

Developed syllabus with primary instructors serving 23 x 2 international high school students. Graded and corrected coursework comprising python code, laboratory reports, problem sets, and research presentations. Provided classroom, laboratory, fieldwork, and homework support.

The University of Chicago, ASTR 12600 Matter, Energy, Space, and Time | College level

Tutor, Autumn 2023

Privately tutored first-year astrophysics student; provided supplementary instruction to address learning gaps and enhance understanding of introductory astrophysics material.

OTHER EMPLOYMENT, ENGAGEMENT, AND STUDENT LIFE

Center for Identity + Inclusion, Emerging Minds Project Facilitator September 2023 – Present
Planned, presented, and facilitated fortnightly panels and group discussions for audiences of 10-30 people on contemporary social issues/identities.

UChicago Transit Explorers and Enthusiasts, Club member 2024 – Present
Participate in public transport exploration trips throughout the city of Chicago, organise fundraising, outreach efforts, and political advocacy for healthy and affordable public transport systems.

Department of Physics, Office Assistant January 2023 – September 2025
Designed internal & external communications materials. Supported staff, graduate, and undergraduate events and recruitment campaigns.

Dougan-Niklason House, Residential dorm Co-president 2023 – 2024
Planned and hosted weekly community events for residential house of over 100 members including local sports, restaurant, and museum outings, as well as community outreach efforts serving local food banks. Fundraised over \$2000 for house activities.

PanAsia, Board member 2023 – 2024
Organised teach-ins on a variety of social and political topics to engage the Asian student community in diversity and equity efforts.

TECHNICAL SKILLS

Coding languages: Python, LaTeX

Computational techniques: N-body simulation, Optimisation, Monte Carlo methods, Artificial neural networks, Convolutional neural networks, Astrophysical Imaging & Spectroscopy, Chemical Abundance Analysis

Astrophysics software: SAO DS9, TOPCAT, LESSPayne, GNU Radio

LANGUAGES

English: Native/Bilingual fluency

Vietnamese: Native/Bilingual fluency

French: Conversational proficiency